OneRoot

AgriRadius Pro

Technical Overview

Satellite intelligence for farm-to-business sourcing.
How we use open satellite maps, radar and public data to
compute land, crops, soil, water, prices - and predict what matters.

OneRoot (ENP Farms Private Limited) | oneroot.farm
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What AgriRadius Pro is

OneRoot AgriRadius Pro turns any location and radius in India into a complete agricultural intelligence report - farmland extent, what grows there and when, soil, water, weather, market prices and a ranked shortlist of villages - computed on demand from free and open satellite and geospatial data.

It is a decision-support tool for sourcing: which belts to buy from, when harvests are coming, how dependable an area's production is, and where prices are best. Every estimate is delivered with an honest confidence level, and it grows more accurate for a region as field staff record what they actually see on the ground.

The problem it solves

- Sourcing decisions today are made with little visibility - which villages grow turmeric, coconut or banana, and how much, is guesswork until someone drives there.
- Physical surveys of a new belt cost days of travel; scaling to new districts multiplies the cost.
- Droughts, crop stress and price swings are discovered after they hit the market - too late to plan around.

AgriRadius Pro replaces the guesswork with a repeatable, satellite-based read of any area in about a minute, verified against ground reality over time.

How it works: the data pipeline

Every analysis follows the same four-stage flow, executed in Google Earth Engine's planetary-scale cloud - no imagery is downloaded or stored locally.

1 - Define area

The user picks a point and radius (or a village). The app builds the buffer and finds every village polygon inside it from Census-2011 boundary datasets.

2 - Gather satellite & open data

For the chosen year the app pulls Sentinel-2 optical imagery, Sentinel-1 radar, Google Dynamic World land cover, rainfall, soil and climate layers over the area.

3 - Compute features & models

It derives vegetation indices, full-year phenology, radar canopy texture and land-cover statistics, then runs detection and classification logic on them.

4 - Present & verify

Results are shown as map layers, tables, charts and reports - each with a confidence level - and can be scored against field observations to calibrate the models.

Because the heavy computation runs in Earth Engine's cloud, the app needs only a browser and works the same on desktop and mobile.

Data sources - all free and open

The app fuses eleven public datasets. The advantage is not the data (anyone can access it) but the combination, the methods, and the ground-truth calibration only OneRoot can run.

Source	What it is	What we derive
Sentinel-2 (ESA)	10 m optical imagery, every ~5 days, red-edge bands	NDVI/red-edge indices, crop cycles, greenness
Sentinel-1 (ESA)	10 m C-band radar; sees through cloud	Paddy flooding, plantation canopy texture
Google Dynamic World	AI land cover from Sentinel-2, near real-time	Land-cover classes and cropland extent
ESA WorldCover / WorldCereal	Independent 10 m land-cover / cropland maps	Cross-checks against Dynamic World
SoilGrids (ISRIC)	Global soil properties, 250 m, from lab samples	pH, organic carbon, nitrogen, CEC, texture
CHIRPS (UCSB)	40+ years rainfall, ~5 km	Rainfall reliability, drought risk
ERA5-Land (ECMWF)	Reanalysis climate	Soil temperature and moisture
Open-Meteo	16-day weather forecast, free API	Dry-window / harvest timing
AGMARKNET (data.gov.in)	Daily APMC mandi prices	Live market prices per commodity
SRTM (NASA)	30 m elevation	Slope / terrain filtering
Census-2011 boundaries	Village / taluk / district polygons	Per-village analysis

Methods: how results are computed

Vegetation & crop cycles

For each month of the year the app computes NDVI (overall greenness) and NDRE (red-edge, which separates crop types far better than NDVI) over cropland only. Fitting the shape of this year-long curve - number of peaks, timing, how long it stays green - classifies the cropping pattern (single, double, long-duration, perennial) and estimates sowing/harvest timing.

Paddy detection - radar

Sentinel-1 radar sees flooded fields through monsoon cloud. Pixels that flood (very low backscatter) then show strong canopy growth over cropland are classified as paddy.

Plantation detection - radar canopy structure

Coconut/arecanut/banana are identified by three independent signals: they stay green all year (high 15th-percentile NDVI, low seasonal swing), their canopy is structurally rough to radar (high Sentinel-1 texture, unlike smooth grass or bare land), and they sit on flat terrain. Combining structure with greenness is what separates a real grove from flat green pasture.

Soil, rainfall & climate

Soil properties are read from SoilGrids at root-zone depth and summarised per village. Rainfall reliability is derived from 40 years of CHIRPS (year-to-year variability); soil temperature/moisture from ERA5-Land; the 16-day forecast from Open-Meteo, which also flags the longest dry window for harvest and transport.

Village sourcing score

Each village is scored 0-100 by combining cropland size, cropping intensity, soil quality, rainfall reliability and field verification - turning eight separate analyses into one ranked shortlist.

What the app produces

- **Map layers** - Dynamic World land cover, cropland-confidence agreement, paddy, plantations, painted soil pH / organic carbon / nitrogen, WorldCereal cropland, village boundaries - all with adjustable transparency.
- **Land-cover statistics** in acres, with a two-dataset confidence range and a 3-year stability check.
- **Crop-cycle analysis -** pattern, cycles per year, growth-peak months, NDVI time-series chart.
- **Paddy and plantation acreage** and share of cropland.
- **Soil profile** per buffer and per village; soil temperature and moisture.
- **Rainfall reliability** (10-year) and 16-day weather outlook with dry-window detection.
- **Live mandi prices** per commodity with the best market highlighted.
- **Per-village insights and a 0-100 sourcing score -** a ranked shortlist of where to focus.
- **One-click PDF and Excel reports** bundling every analysis for an area.

Accuracy, trust and calibration

The app is deliberately honest about certainty. It separates what is measured from what is modelled, and it improves itself over time.

Confidence tiers

- **High** - NDVI/crop vigour, rainfall, radar paddy, mandi prices, forecast: direct measurements, not models.
- **Use as ranges** - land-cover classes (~75-85% accuracy), soil maps (250 m modelled), pattern thresholds.

Built-in safeguards

- Two independent datasets (Dynamic World + ESA WorldCover/WorldCereal) are cross-checked; agreement is reported.
- A 3-year stability test flags landscapes where the classification is unreliable.
- Terrain and radar-structure filters remove common false positives.
- Every figure is presented as a range where appropriate, not false precision.

The calibration flywheel

Field staff log what they actually see (crops, cycles, soil-card values) against each village. The app scores its predictions against these observations and reports a live accuracy figure. Once enough labelled villages accumulate in a region, a Random Forest classifier is trained on those real labels using the full feature stack (red-edge + phenology + radar) - producing a crop map tuned to that region, better than any global product. Predict -> verify -> retrain: the system gets sharper every week, and the labelled dataset becomes an asset no competitor can buy.

Technology & deployment

- **Compute:** Google Earth Engine (planetary-scale, free for this use) - no local imagery storage.
- **App:** Python + Streamlit; interactive Leaflet maps; runs in any browser, desktop or mobile.
- **Field data:** ground-truth and soil cards saved to a shared Google Sheet, so corrections persist centrally.
- **Hosting:** deployable to Streamlit Community Cloud with a permanent URL - runs 24/7, no dedicated server.
- **Security:** a single Google service account powers Earth Engine and Sheets; secrets are never committed.

Roadmap

Phase	Focus
Now	Desktop use; staff-entered ground truth; internal testing across areas
Next	Cloud deployment; field-staff mobile access; crop-stress early warning; harvest-timing tracker
Later	Region-tuned trained crop classifier; measured NPK & groundwater maps from field cards; supply forecasts weeks ahead of the market

Built with honest confidence levels and verified against ground reality - a tool you can trust the direction of today, and improve into precision over time.